Hallucination and Knowledge-Cutoff Probe for a Library Assistant

Literature Survey

1. Literature Survey

2.1 Hallucination in language models. Ji et al. [1] provide the foundational survey of hallucination in natural language generation and distinguish intrinsic hallucinations, which contradict the input, from extrinsic ones, which cannot be verified from it. Huang et al. [2] and Zhang et al. [3] update this for LLMs, organising causes (data, training, inference) and mitigations, and note that fact-conflicting hallucination is the most consequential for real-world use. TruthfulQA [4] shows that models can reproduce popular misconceptions and that scale alone does not remove the problem, motivating adversarial probes built on false premises.

2.2 Detecting and measuring hallucination. SelfCheckGPT [5] detects hallucination without external resources by sampling several responses and measuring their consistency, since fabricated facts tend to vary between samples. FActScore [6] breaks long text into atomic facts and computes the proportion supported by a knowledge source, an approach that fits catalogue records well. Kadavath et al. [7] show that models can, to a useful degree, estimate the correctness of their own answers, supporting the idea of abstention when confidence is low. Mallen et al. [8] show that accuracy on rare entities is poor for parametric memory, which is directly relevant because most catalogue items are long-tail.

2.3 Knowledge cutoff and temporal generalisation. Lazaridou et al. [9] show that models trained on past data perform worse on future text, and Dhingra et al. [10] propose explicitly time-aware models. RealTime QA [11] provides a benchmark of questions whose answers change, and FreshLLMs [12] shows that search-engine augmentation helps with fast-changing knowledge. Cheng et al. [13] find that a model's effective cutoff can differ from its stated one, which motivates measuring the effective cutoff empirically rather than trusting documentation.

2.4 Retrieval grounding. Retrieval-augmented generation [14] couples a generator with an external document store and is the dominant approach for knowledge-intensive tasks; Gao et al. [15] survey retriever, generator and augmentation design choices. For libraries, the catalogue, policy pages and new-acquisition feeds form a natural retrievable corpus.

2.5 Summary of the literature.

Ref.	Work	Focus	Relevance to this project
[1]	Ji et al., 2023	Survey of hallucination in NLG; intrinsic vs extrinsic	Basis for the hallucination taxonomy
[2]	Huang et al., 2023	LLM hallucination survey: causes, detection, mitigation	Categories and mitigation overview
[3]	Zhang et al., 2023	Survey of hallucination in LLMs	Complementary taxonomy and causes
[4]	Lin et al., 2022 (TruthfulQA)	Benchmark of imitative falsehoods	Model for adversarial probe design
[5]	Manakul et al., 2023 (SelfCheckGPT)	Sampling-consistency hallucination detection	Verifier for free-text claims
[6]	Min et al., 2023 (FActScore)	Atomic-claim factual precision	Annotation and scoring unit
[7]	Kadavath et al., 2022	Models partially know what they know	Abstention and uncertainty signals
[8]	Mallen et al., 2023	Failure on long-tail entities; retrieval helps	Why catalogue items are high-risk
[9]	Lazaridou et al., 2021	Temporal generalisation degradation	Basis for knowledge-cutoff probe
[10]	Dhingra et al., 2022	Time-aware language models	Temporal knowledge modelling

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[11]	Kasai et al., 2023 (RealTime QA)	QA on changing facts	Design of time-sensitive probes
[12]	Vu et al., 2023 (FreshLLMs)	Search augmentation for fresh knowledge	Retrieval for post-cutoff queries
[13]	Cheng et al., 2024	Effective vs reported cutoffs	Nominal vs effective cutoff metric
[14]	Lewis et al., 2020	Retrieval-augmented generation	Core grounding architecture
[15]	Gao et al., 2023	RAG survey	Design choices for retriever and generator

2. Research Gap

The surveyed work addresses hallucination and temporal staleness mainly with general-purpose benchmarks on encyclopaedic facts. We did not find a probe designed specifically for library assistants that combines (i) bibliographic hallucination types such as phantom items and fabricated identifiers, (ii) library-specific time-sensitive information such as editions and policy changes, and (iii) catalogue-derived ground truth. This project targets that gap, as reflected in Papers 1-3.

References

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